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B.Tech. Degree IV Semester Examination April 2017

ME 15-1402 METROLOGY AND INSTRUMENTATION

(2015 Scheme)

Time: 3 Hours

Maximum Marks:60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

- I. (a) Differentiate between "Hole basis system" and "Shaft basis system".
 (b) What is the difference between standard gauges and limit gauges?
 (c) Calculate the gauge block buildup required to set a 10 cm sine bar to an angle of 30 degree.
 (d) Explain the formation of white and dark fringes due to interference of light.
 (e) The magnification of a projector is 25X. If the screen diameter is 1 m, what is the maximum diameter of the job, which is possible?
 (f) Draw the functional block diagram of a measurement system.
 (g) What is meant by static calibration?
 (h) Explain the constructional details of RTD.
 (i) What is a strain gage rosette?
 (j) Explain the term strain sensitivity.

PART B

(4 × 10 = 40)

- II. (a) Explain the terms "Cosine error" and "Parallax error" with suitable sketches. (5)
 (b) In a hole and shaft assembly of 30 mm nominal size, the tolerances for hole and shaft are specified as Hole: $30^{+0.02}_{-0.00}$ mm, Shaft $30^{+0.40}_{-0.70}$ mm. Determine: (3)
 (i) Maximum and minimum clearance obtainable.
 (ii) Hole and shaft tolerance.
 (iii) The type of fit.
 (c) What are the functions of the following gauges: (2)
 (i) Form gauge (ii) Reference gauge (iii) Feeler gauge (iv) Air gauge.

OR

- III. (a) Distinguish between "Line standard" and "End standard" with examples. (8)
 (b) What do you mean by the 'accuracy of the instrument is better than 2%'? (2)

- IV. (a) Describe the principle, working and uses of a Clinometer. (5)
 (b) Define the following in connection with surface texture assessment. (5)
 (i) Roughness (ii) Waviness (iii) Lay (iv) Sampling length.

OR

- V. (a) Describe and discuss the working principle and application of a Toolmaker's microscope. (8)
 (b) Why is sine bar not preferred for measuring angle more than 45°? (2)

(P.T.O.)

VI. Discuss briefly the generalized input-output configuration of a measurement system. (10)

OR

- VII. (a) Define the following terms. (5)
(i) Sensitivity (ii) Hysteresis.
(b) Describe the working of potentiometer transducer as a zero-order instrument. (5)

- VIII. (a) Explain the working of optical pyrometer. (4)
(b) Explain the following. (6)
(i) Gas chromatography (ii) Scintillation counter.

OR

- IX. (a) With a neat sketch explain Heenan and Froude hydraulic dynamometer. (6)
(b) Explain microphone for acoustic measurement. (4)